

BCSE 0201: BACKEND ENGINEERING

Objective: To enable students to design and develop scalable full-stack web applications using Node.js, Express, MongoDB, Git, RESTful APIs, authentication, and deployment practices, with a strong focus on real-world backend development and security.

Prerequisite: Object-Oriented Programming, Database Management Systems

Credits: 3

L-T-P: 3-0-2

Module No.	Content	Teaching Hours
I	Unix Commands and Git and GitHub: Exploring Unix environment and basic Unix commands, Learning all about VCS and why we need them, Exploring git in detail and learning how to commit changes, resolve conflicts, etc., Learning all about GitHub. First Brush with NodeJS : Introducing Node.js and Learning all about Node, how its is useful, what we can do using Node etc., Learning various inbuilt Node modules e.g. : http, file-system, path etc., Sharing code between JS files in Node.js, Working with file system, Learning how to read and write data into the files, Introducing out Web Framework for creating server, Comparing library and frameworks, Creating our first express app, Learning all about routing , Using nodemon as a tool to restart server automatically, Understanding templating languages and why do we need them, Configuring express to use various templating engine such as EJS, Using stylesheets and scripts in our express app as static files. RESTfull Routing: Introducing GET vs POST request and where do we use them, Understanding POST request in detail, Parsing the request body using express body parsing middleware, Learning all about REST architecture and creating REST routes for CRUD app.	21
II	Introduction Databases: Introducing databases as persistent layer, Understanding SQL and NoSQL databases, Learning how to select databases for different projects using CAP theorem, Learning all about MongoDB how it stores data etc., understanding difference between JSON and BSON, Exploring various mongo queries for inserting, deleting, updating data in collections, Introducing mongoose as ODM, Creating schema using mongoose and using Model to manipulate the data, Interacting with MongoDB using mongoose and performing - Insert, delete, update, Read operations. Middleware, Express Session and Cookies : Learning all about middlewares and why do we need them, Exploring how to write Express middleware , protecting our routes with the help of middleware, Exploring mongoose middleware, Learning all about cookies -Client Side Storage, Exploring express session and how it helps and in storing data on server side, Diving into authentication and understanding Authentication vs Authorization, Learning about Cryptographic Hashing Functions, Implementing authentication using Bcrypt from scratch, Using sessions for persistent login, Protecting routes using middleware, Learning about passport.js and adding Authentication using passport, Adding authorization using middleware's in our app Common Security Issues and Deployment : Understanding common security issues using and Mongo Injection, Cross site scripting and how to mitigate small attacks, Sanitizing payload before storing the data into the database, Using mongo store for our sessions, Using helmet to prevent various attacks, Preparing and deploying our app to production environment, Connecting our App with cloud database	21

	using MongoDB Atlas, Deploying and Monitoring our app to Heroku.	
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Text Book:

- **Node.JS Web Development by David Herron**

Reference Books:

- **Eloquent JavaScript : A Modern Introduction to Programming**
Book by Marijn Haverbeke
- **You Don't Know JS**
Book by Kyle Simpson

Focus: This Course focuses on Employability under CO2, CO3, and CO4.

Outcome:

After successful completion of this course, the students will be able to:

- CO1: Demonstrate proficiency in Unix commands and Git for version control and collaborative development. CO2: Develop server-side applications using Node.js and Express, including routing, templating, and middleware usage. CO3: Implement RESTful APIs and handle CRUD operations with Express and MongoDB using Mongoose ODM. CO4: Apply authentication, authorization, sessions, cookies, and middleware to secure Node.js web applications.